# Software Requirements Specification

Student Record System

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## Revision History

| Version | Date | Author | Change Summary | Approval |
|---|---|---|---|---|
| **0.1** | 13-09-2026 | Tanmay T A | Initial SRS draft containing legacy extended requirements, initial analysis models, and appendices A–D. | Draft |

| Version | Date | Author | Change Summary | Approval |
|---|---|---|---|---|
| **0.2** | 27-09-2026 | Uttam B S | Restructured SRS to align with standard SE template; removed out-of-scope appendices, streamlined NFR targets, eliminated complex OTP/geolocation constraints, and introduced the 48-hour attendance freeze rule. | Reviewed |
| **0.3** | 06-10-2026 | Vibha Vasisht | Refined functional requirements: introduced student self-enrollment (max 6 courses), administrative user provisioning, restricted administrative editing of personal contact data, and defined the 4-stage grade lifecycle (`Draft` → `Submitted` → `Under Review` → `Published`). | Reviewed |

| Version | Date | Author | Change Summary | Approval |
|---|---|---|---|---|
| **1.0** | 09-10-2026 | Tanmay T A | Finalized requirement specifications (20 FRs, 5 NFRs, 2 PRJ-SO, 5 PRJ-SR), added dynamic rubric weight auto-normalization, and structured split UML Use-Case models for Academic Operations and System Administration. | Approved |

## Approvals

| Role | Name | Signature / Email | Date |
|---|---|---|---|
| Course Instructor | Dr. Adithya Balasubramanyam | adithyab@pes.edu | 13-10-2026 |

# 1 Introduction

## 1.1 Purpose

This document specifies the Software Requirements Specification (SRS) for the Student Record System. It details the functional requirements, non-functional constraints, security objectives, and system models necessary for university-level deployment. This document serves as the primary technical reference for software developers, system evaluators, and academic instructors during project implementation and acceptance testing.

## 1.2 Scope

The system provides digitized academic management capabilities covering administrative account provisioning, course section registration with weighted evaluation rubrics, student self-enrollment, digitized roll-call attendance marking with a strict 48-hour freeze constraint, and a multi-stage grade evaluation workflow (`Draft` → `Submitted` → `Under Review` → `Published`).

The scope explicitly excludes automated mail distribution (credentials are displayed on-screen during administrative account creation), physical hardware card integration, and external payment gateway processing.

### 1.3 Audience

This document is targeted toward:

- **Software Engineering Faculty & Evaluators:** To review architecture compliance, constraint satisfaction, and acceptance criteria.
- **Development Team (3 Students):** To guide API schema design, backend logic, and frontend role-based workflows during 2 one-week Agile sprints.
- **System Administrators & Users:** To understand user role capabilities, system boundaries, and operational constraints.

### 1.4 Definitions and Acronyms

| Term / Acronym | Definition |
|---|---|
| **API** | Application Programming Interface |
| **GPA** | Grade Point Average |
| **JWT** | JSON Web Token (used for secure session authorization) |
| **NFR** | Non-Functional Requirement |
| **ORM** | Object-Relational Mapping |
| **RBAC** | Role-Based Access Control |
| **RTM** | Requirements Traceability Matrix |
| **SMART** | Specific, Measurable, Achievable, Relevant, Time-bound criteria |
| **SRS** | Software Requirements Specification |
| **TLS** | Transport Layer Security (version 1.3) |

## 2 Overall Description

### 2.1 Product Perspective

The Academic Management & Record System is an independent, responsive web service designed to replace paper-based academic tracking across a higher education institution of approximately 10,000 students and staff. It operates in a standard 3-tier web architecture consisting of a modern single-page client interface, a RESTful API application server with middleware role-based access control, and a relational SQL database store.

### 2.2 Major Product Functions

- **Account Provisioning & Profile Management:** Administrative creation of Student, Faculty, and Admin accounts; mandatory password changes upon first login; self-service contact detail updates.

- **Course Offering & Self-Enrollment:** Administrative course registration, assignment of primary faculty, setting evaluation rubrics (with backend auto-normalization to 100%), and student self-enrollment (capped at 6 courses per term).
- **Schedule & Roll-Call Attendance:** Class schedule publishing; digitized instructor roll-call; 48-hour automated attendance freeze; administrative override capability.
- **Weighted Rubric Grading Workflow:** Faculty mark entry against weighted components; transition through state lifecycle (`Draft` → `Submitted` → `Published`); student regrade request dispute workflow (`Under Review`).
- **Academic Dashboards:** Real-time student GPA and attendance metrics; class grade distribution summaries; administrative institution-level reporting.

### 2.3 User Roles and Characteristics

| Role | Characteristics & Responsibilities |
|---|---|
| **Student** | Enrolls in up to 6 active courses during registration windows; views session-by-session attendance and grade distributions; submits regrade dispute requests; updates personal contact information. |
| **Faculty** | Conducts class roll-calls for assigned sessions; updates attendance within 48 hours; inputs rubric marks; submits draft grades for administrative review; resolves student regrade requests. |
| **Administrator** | Provisions user accounts; creates courses and schedule sessions; sets evaluation rubrics; publishes finalized grades; executes administrative attendance overrides; manages account operational states. |

### 2.4 Operating Environment

- **Server Environment:** Linux container execution environment (Docker) running Python application services and MySQL relational database.
- **Client Environment:** Responsive web browser (Google Chrome, Mozilla Firefox, Apple Safari, Microsoft Edge) operating on devices with screen widths from 375px to 1920px.
- **Network Infrastructure:** Standard institutional WAN/LAN with mandatory HTTPS (TLS 1.3) traffic routing.

### 2.5 Constraints

- **Agile Sprint Scope:** Implementation is bounded by 2 one-week-long Agile sprints delivered by a 3-student engineering team.
- **Zero Mail Automation:** External SMTP server setup is excluded; temporary credentials must be rendered directly on administrative confirmation UI views.

- **Hard 48-Hour Attendance Freeze:** Faculty attendance edit permissions are strictly revoked 48 hours post-session by automated backend time checks.
- **Course Limits:** Student enrollment is capped at a strict upper limit of 6 courses per academic term.

# 3 External Interface Requirements

## 3.1 User Interfaces

- **General UI Layout:** Responsive web interface built using modern CSS layout standards. Core workflows (e.g., taking roll-call, checking GPA) must be accessible within 3 UI clicks from login.
- **Student Dashboard:** Single-screen view rendering enrolled course cards, GPA summary, aggregate attendance percentages, and grade state indicators.
- **Faculty Marking Interface:** Tabular class roster with toggle inputs for binary attendance (Present/Absent) and numerical grade input fields per rubric component.
- **Admin Control Console:** Form interfaces for provisioning user accounts, uploading class schedules, setting course rubric weights, and executing grade state updates.

## 3.2 Hardware Interfaces

- The system requires no direct hardware access or custom peripheral drivers. Standard client display devices, keyboards, and pointing devices (touchscreen or mouse) are supported over standard web protocols.

## 3.3 Software Interfaces

- **Database Management System:** Relational SQL database connected via Object-Relational Mapping (ORM) client drivers over encrypted connections.
- **Authentication Engine:** HTTP-only JSON Web Token (JWT) handling session state and role claims.

## 3.4 Communication Interfaces

- **Protocol:** HTTP/2 over TLS 1.3 (HTTPS) enforced across all endpoints. Plain HTTP requests automatically redirect to HTTPS with HSTS headers enabled.
- **Data Format:** Restful JSON payloads for all API requests and responses.

# 4 System Features (Detailed Functional Requirements)

## 4.1 Module 1: Account Provisioning & User Profile

| Req ID | Requirement Statement | Type | Priority | Source / Stakeholder | Acceptance Criteria / Test Ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| **SYS-F-001** | The system shall allow Administrators to provision user accounts with assigned roles (Student, Faculty, Admin) and display initial temporary login credentials on a dedicated confirmation screen upon creation within 2 seconds. | Functional | High | Admin | AC-SYS-F-001: Admin creates account; temporary credentials display on screen once. Test: TC-ACC-01 | Requires Admin RBAC privilege. |
| **SYS-F-002** | The system shall mandate users logging in with temporary credentials to update their account password before granting access to application features. | Functional | High | Security | AC-SYS-F-002: Login with temporary password forces redirect to password change screen. Test: TC-ACC-02 | Depends on SYS-F-001. |

| Req ID | Requirement Statement | Type | Priority | Source / Stakeholder | Acceptance Criteria / Test Ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| **SYS-F-003** | The system shall allow logged-in Students and Faculty to view and update their personal contact details (phone number, home address, emergency contact) at any time. | Functional | Medium | Student / Faculty | AC-SYS-F-003: Profile updates persist in database and display immediately upon saving. Test: TC-ACC-03 | Accessible by account owner. |
| **SYS-F-004** | The system shall allow Administrators to view user account metadata, reassign roles, or deactivate user accounts across the institution. | Functional | High | Admin | AC-SYS-F-004: Deactivated accounts are blocked from logging in within 1 second. Test: TC-ACC-04 | Admin privilege required. |

### 4.2 Module 2: Course Setup, Self-Enrollment & Schedule Management

| Req ID | Requirement Statement | Type | Priority | Source / Stakeholder | Acceptance Criteria / Test Ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| **SYS-F-005** | The system shall allow Administrators to register course offerings, set section capacity limits, assign primary faculty, and set weighted rubric evaluation schemes with backend auto-normalization to 100%. | Functional | High | Admin / Faculty | AC-SYS-F-005: Course created with assigned faculty; raw rubric weights normalize to 100% automatically. Test: TC-CRS-01 | Baseline for grading. |
| **SYS-F-006** | The system shall allow Students to self-enroll in up to 6 active course sections during an open enrollment window set by the Administrator. | Functional | High | Student | AC-SYS-F-006: Student can enroll up to 6 courses; 7th attempt or full section attempt is rejected. Test: TC-CRS-02 | Requires active registration window. |

| Req ID | Requirement Statement | Type | Priority | Source / Stakeholder | Acceptance Criteria / Test Ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| **SYS-F-007** | The system shall allow Administrators to create class schedule sessions defining course section, date, start/end time, room, and assigned instructor. | Functional | High | Admin | AC-SYS-F-007: Created schedule populates calendar views for instructor and enrolled students. Test: TC-CRS-03 | Requires registered course sections. |
| **SYS-F-008** | The system shall restrict Faculty roster visibility and session access to only those course sections assigned to them by the Administrator. | Functional | High | Faculty | AC-SYS-F-008: Faculty dashboard displays only assigned sections; unauthorized access returns HTTP 403. Test: TC-CRS-04 | RBAC enforcement. |

### 4.3 Module 3: Digitized Attendance Marking & Freeze Constraints

| Req ID | Requirement Statement | Type | Priority | Source / Stakeholder | Acceptance Criteria / Test Ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| **SYS-F-009** | The system shall allow Faculty to mark binary attendance (Present/Absent) for enrolled students during scheduled sessions assigned to them. | Functional | High | Faculty | AC-SYS-F-009: Attendance submission saves entries for enrolled students in valid assigned session. Test: TC-ATT-01 | Depends on SYS-F-007, SYS-F-008. |
| **SYS-F-010** | The system shall automatically lock attendance modification rights for Faculty exactly 48 hours after a session's scheduled start time. | Functional | High | Operations | AC-SYS-F-010: Faculty edit controls disable 48h post-session; backend rejects edits post-48h. Test: TC-ATT-02 | Automated timestamp validation. |
| **SYS-F-011** | The system shall allow Administrators to override and modify locked attendance records beyond the 48-hour window. | Functional | Medium | Admin | AC-SYS-F-011: Admin updates locked attendance record with administrative audit record. Test: TC-ATT-03 | Admin privilege required. |

| Req ID | Requirement Statement | Type | Priority | Source / Stakeholder | Acceptance Criteria / Test Ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| **SYS-F-012** | The system shall display aggregate and session-by-session attendance percentages to Students for all currently enrolled courses. | Functional | High | Student | AC-SYS-F-012: Attendance percentages update within 1s of Faculty submission. Test: TC-ATT-04 | Depends on SYS-F-009. |

### 4.4 Module 4: Weighted Rubric Grading & Grade Lifecycle

| Req ID | Requirement Statement | Type | Priority | Source / Stakeholder | Acceptance Criteria / Test Ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| **SYS-F-013** | The system shall allow Faculty to record numeric marks against predefined rubric components for enrolled students, saving initial entries in `Draft` state. | Functional | High | Faculty | AC-SYS-F-013: Marks saved in `Draft` state remain hidden from student accounts. Test: TC-GRD-01 | Depends on SYS-F-005. |

| Req ID | Requirement Statement | Type | Priority | Source / Stakeholder | Acceptance Criteria / Test Ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| **SYS-F-014** | The system shall allow Faculty to submit draft grades, calculating total course scores based on normalized rubric weights and transitioning state to `Submitted`. | Functional | High | Faculty | AC-SYS-F-014: Total score calculates accurately; grade state transitions to `Submitted`. Test: TC-GRD-02 | Depends on SYS-F-013. |
| **SYS-F-015** | The system shall allow Administrators to publish `Submitted` grades, changing state to `Published` and rendering final grades visible on student dashboards. | Functional | High | Admin / Student | AC-SYS-F-015: Grade state updates to `Published`; grades display on student dashboard. Test: TC-GRD-03 | Admin approval step. |

| Req ID | Requirement Statement | Type | Priority | Source / Stakeholder | Acceptance Criteria / Test Ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| **SYS-F-016** | The system shall allow Students to submit regrade requests for `Published` grades, transitioning the grade state to `Under Review`. | Functional | Medium | Student | AC-SYS-F-016: Dispute text attaches; state updates to `Under Review` on Faculty/Admin view. Test: TC-GRD-04 | Student challenge workflow. |
| **SYS-F-017** | The system shall allow Faculty or Admin to resolve `Under Review` disputes by adjusting rubric marks and re-publishing the finalized grade. | Functional | Medium | Faculty / Admin | AC-SYS-F-017: Rubric marks recalculate; state transitions back to `Published`. Test: TC-GRD-05 | Resolution workflow. |

### 4.5 Module 5: Analytics & Summary Reporting

| Req ID | Requirement Statement | Type | Priority | Source / Stakeholder | Acceptance Criteria / Test Ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| **SYS-F-018** | The system shall render a consolidated academic dashboard for Students summarizing overall GPA, individual course grades, and attendance rates within 800 ms. | Functional | High | Student | AC-SYS-F-018: Student dashboard loads current GPA, grades, and attendance within 800ms. Test: TC-REP-01 | Consolidated view. |
| **SYS-F-019** | The system shall generate class summary reports for Faculty displaying grade distributions and attendance averages per assigned section. | Functional | Medium | Faculty | AC-SYS-F-019: Class summary view displays section mean, median, and attendance rate. Test: TC-REP-02 | Faculty analytics. |

| Req ID | Requirement Statement | Type | Priority | Source / Stakeholder | Acceptance Criteria / Test Ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| **SYS-F-020** | The system shall allow Administrators to export institution-level summary metrics detailing total enrollments, unsubmitted grades, and attendance rates. | Functional | Medium | Admin | AC-SYS-F-020: Admin view generates institution metrics across active courses within 2s. Test: TC-REP-03 | Executive reporting. |

# 5 Non-Functional & Security Requirements

## 5.1 Non-Functional Requirements

| Req ID | Requirement Statement | Category | Priority | Acceptance Criteria / Measurement |
|---|---|---|---|---|
| **SYS-NF-001** | The backend API shall process 95% of standard HTTP requests with a response time of less than 800 ms under a baseline load of 500 active sessions. | Performance | High | 95th percentile latency $< 800$ ms verified via automated load suite. Test: TC-PERF-01 |

| Req ID | Requirement Statement | Category | Priority | Acceptance Criteria / Measurement |
|---|---|---|---|---|
| **SYS-NF-002** | The database shall process 200 concurrent attendance and grade submission transactions without locking failures or transaction drops. | Concurrency | High | Automated concurrency test executes 200 simultaneous POST requests with 0% failure. Test: TC-PERF-02 |
| **SYS-NF-003** | The application shall maintain 99.0% operational availability during active academic term periods, excluding scheduled maintenance windows. | Availability | High | System uptime monitoring logs show $\geq$ 99.0% monthly uptime. Test: TC-OPS-01 |
| **SYS-NF-004** | The web application interface shall be fully responsive across display viewports from 375px to 1920px width, allowing key workflows within 3 clicks. | Usability | Medium | UI audit passes responsiveness tests; roll-call execution completes in $\leq 3$ clicks. Test: TC-UX-01 |
| **SYS-NF-005** | The software architecture shall strictly separate Presentation, Application, and Data layers with automated unit test coverage of $\geq$ 70%. | Maintainability | High | Automated test coverage tools confirm $\geq$ 70% branch/statement coverage. Test: TC-QUAL-01 |

## 5.2 Security

### 5.2.1 Security Objectives

| ID | Objective Title | Description |
|---|---|---|
| **PRJ-SO-001** | Data Isolation & RBAC | Guarantee complete role-based data isolation so that student academic records, grades, and personal details are strictly confidential and accessible only to authorized users. |
| **PRJ-SO-002** | Workflow State Integrity | Ensure non-repudiation and integrity of academic records by enforcing immutable attendance rules (48-hour lock) and rigid grade state transitions (`Draft` → `Submitted` → `Under Review` → `Published`). |

### 5.2.2 Security Requirements

| Req ID | Requirement Statement | Type | Priority | Source / Stakeholder | Acceptance Criteria / Test Ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| **PRJ-SR-001** | All network communications between client web applications and system backend API services shall enforce TLS 1.3 encryption. | Security | High | Security | AC-PRJ-SR-001: Plain HTTP requests redirect to HTTPS; weak ciphers rejected. Test: TC-SEC-01 | TLS 1.3 configuration. |

| Req ID | Requirement Statement | Type | Priority | Source / Stakeholder | Acceptance Criteria / Test Ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| **PRJ-SR-002** | User account passwords shall be salted and hashed using bcrypt with a minimum work factor of 12 prior to database storage. | Security | High | Security | AC-PRJ-SR-002: Passwords stored strictly as bcrypt hashes in database tables. Test: TC-SEC-02 | Auth Module. |
| **PRJ-SR-003** | User sessions shall be authorized using HTTP-only JSON Web Tokens (JWT) with an automated expiration lifetime of 48 hours. | Security | High | Security | AC-PRJ-SR-003: Requests with tokens older than 48 hours return HTTP 401 Unauthorized. Test: TC-SEC-03 | Auth Middleware. |
| **PRJ-SR-004** | The system shall enforce Role-Based Access Control (RBAC) middleware on every API endpoint to restrict resource access according to user role. | Security | High | Security | AC-PRJ-SR-004: Accessing unauthorized role endpoints returns HTTP 403 Forbidden. Test: TC-SEC-04 | RBAC Middleware. |

| Req ID | Requirement Statement | Type | Priority | Source / Stakeholder | Acceptance Criteria / Test Ref | Comments / Dependencies |
|---|---|---|---|---|---|---|
| **PRJ-SR-005** | All user inputs shall be sanitized and parameterized using Object-Relational Mapping (ORM) queries to prevent SQL Injection and XSS attacks. | Security | High | Security | AC-PRJ-SR-005: Security vulnerability scans confirm 0 SQLi and XSS flaws on input forms. Test: TC-SEC-05 | Input Sanitization. |

# 6 Quality Attributes & Acceptance Tests

## 6.1 Exit Criteria for Acceptance

System acceptance for deployment requires meeting the following criteria: 1. **Functional Completeness:** 100% of high-priority Functional Requirements (`SYS-F-001` through `SYS-F-020`) implemented and verified against acceptance criteria. 2. **NFR Compliance:** Performance response time ($\leq$ 800 ms at 95th percentile) and concurrency (200 simultaneous transactions) validated through automated test scripts. 3. **Security Standards:** Zero high or critical security vulnerabilities detected; pass criteria met for all security requirements (`PRJ-SR-001` to `PRJ-SR-005`). 4. **Traceability Matrix:** 100% test case coverage verified across all requirements in the Requirements Traceability Matrix.

## 6.2 Acceptance Test Suites

| Test Suite ID | Target Module / Area | Primary Test Focus | Target Criteria |
|---|---|---|---|
| **TS-AUTH** | Account Provisioning | User creation, temporary credential display, mandatory password reset. | TC-ACC-01 to TC-ACC-04 Pass |
| **TS-CRS** | Course & Schedule | Registration, student self-enrollment (max 6), roster visibility. | TC-CRS-01 to TC-CRS-04 Pass |

| Test Suite ID | Target Module / Area | Primary Test Focus | Target Criteria |
|---|---|---|---|
| **TS-ATT** | Attendance marking | Roll-call input, 48-hour automated lock, admin override. | TC-ATT-01 to TC-ATT-04 Pass |
| **TS-GRD** | Grade Lifecycle | Rubric calculation, `Draft` $\rightarrow$ `Published` states, regrade dispute. | TC-GRD-01 to TC-GRD-05 Pass |
| **TS-PERF** | System Performance | Latency $\leq$ 800 ms, 200 concurrent submission transactions. | TC-PERF-01, TC-PERF-02 Pass |
| **TS-SEC** | System Security | TLS enforcement, bcrypt hashing, JWT 48h timeout, RBAC check. | TC-SEC-01 to TC-SEC-05 Pass |

# 7 System Models & Diagrams

## 7.1 Architectural Model

The application follows a standard 3-tier architecture separating user interface presentation, server-side business logic, and database persistence.

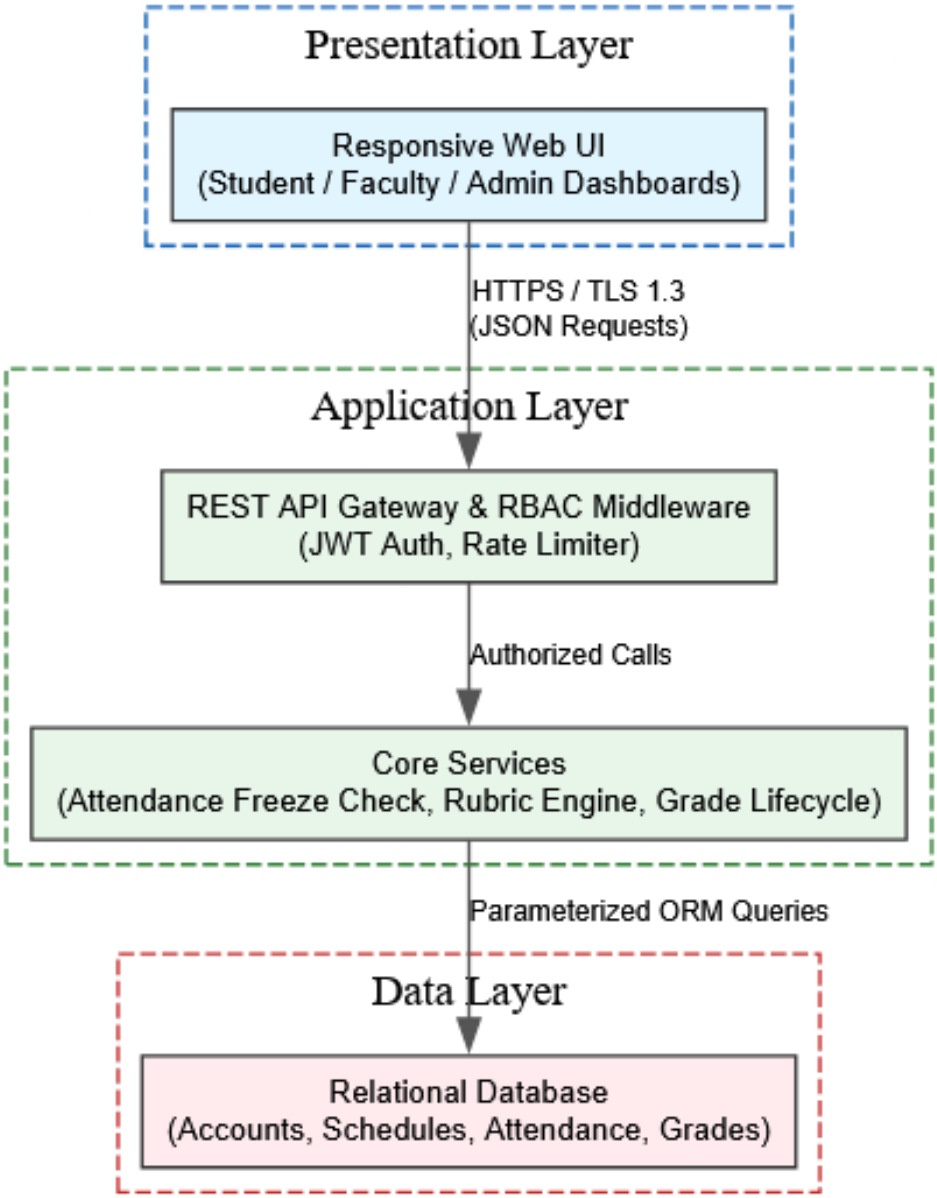



Figure 1: Three-Tier Architecture Model

## 7.2 Data Entity-Relationship Model

The entity-relationship model defines user accounts, course offerings, class schedules, roll-call attendance logs, and weighted rubric grades.

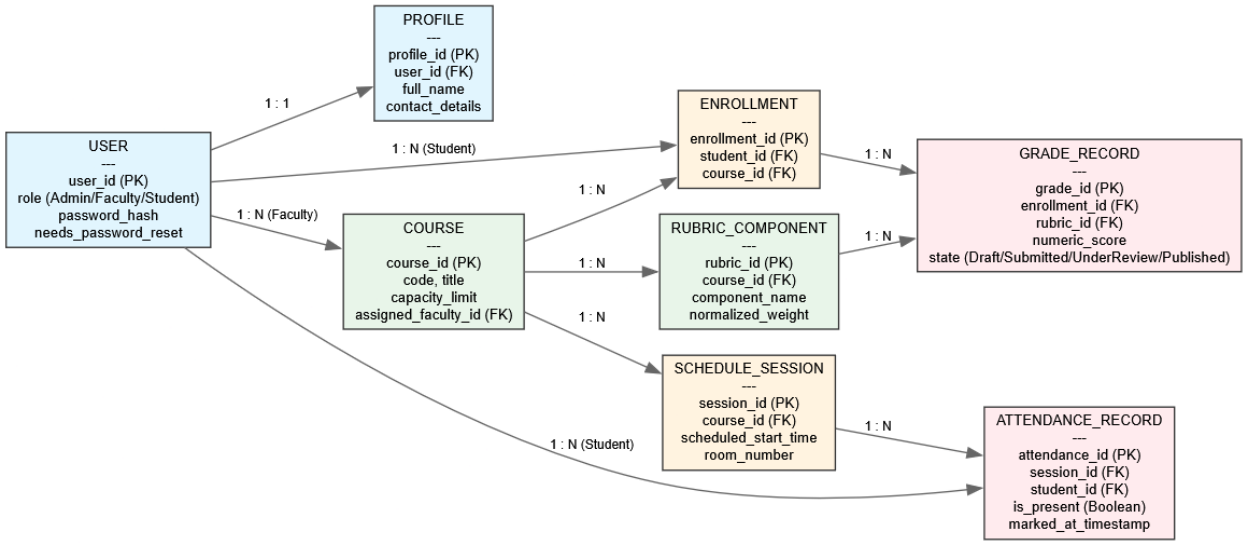



Figure 2: Entity Relationship Diagram

## 7.3 Grade Lifecycle State Machine

Grades progress through a controlled state machine ensuring visibility restrictions and dispute tracking.




Figure 3: Grade Lifecycle State Machine

## 7.4 UML Use-Case Diagrams

### 7.4.1 Academic Operations Use-Case Diagram

Covers primary operational interactions executed by Students and Faculty.

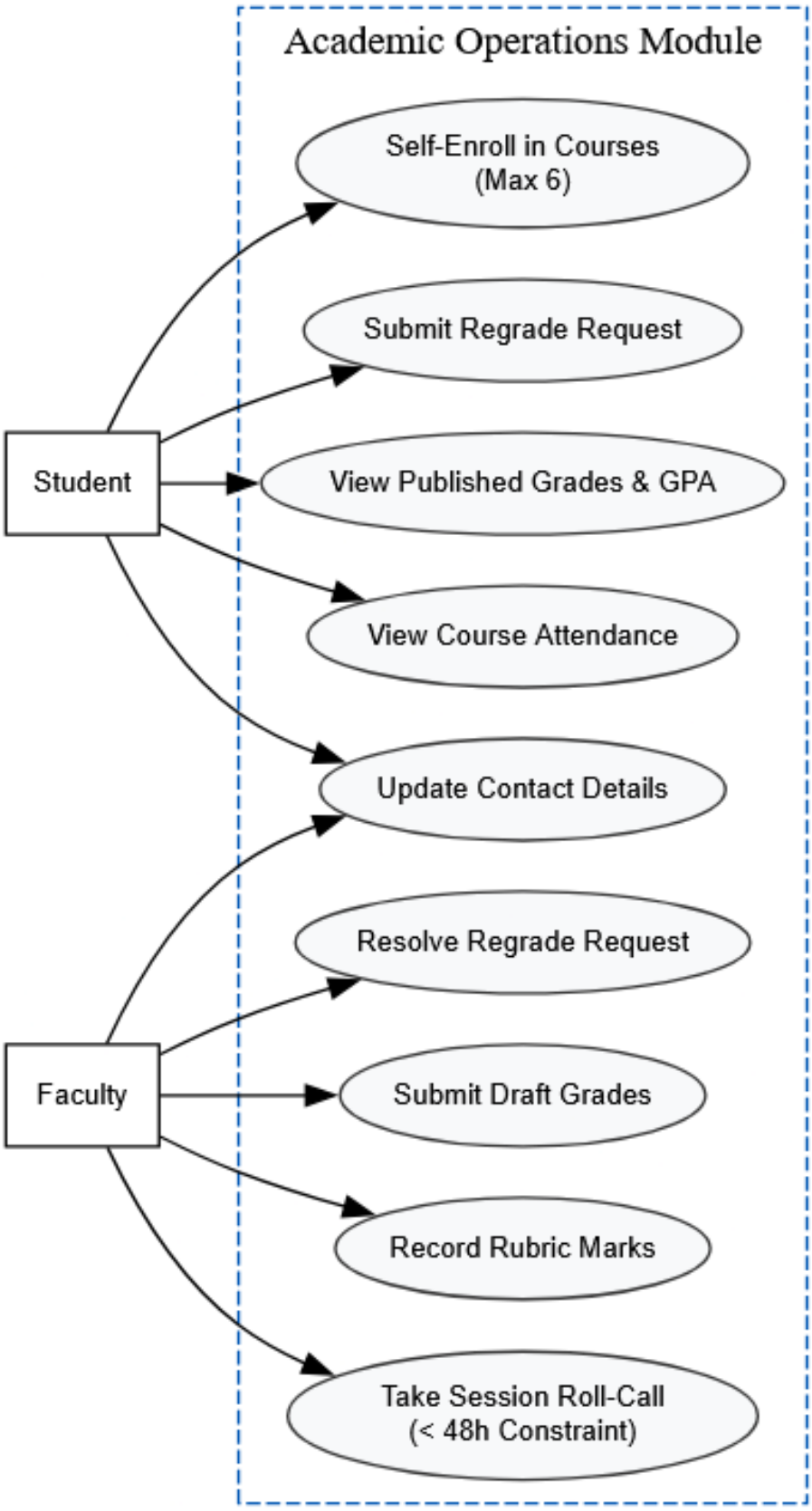



Figure 4: Academic Operations Use Case Diagram

### 7.4.2 System Administration Use-Case Diagram

Covers administrative provisioning, course setup, schedule creation, and grade publication.

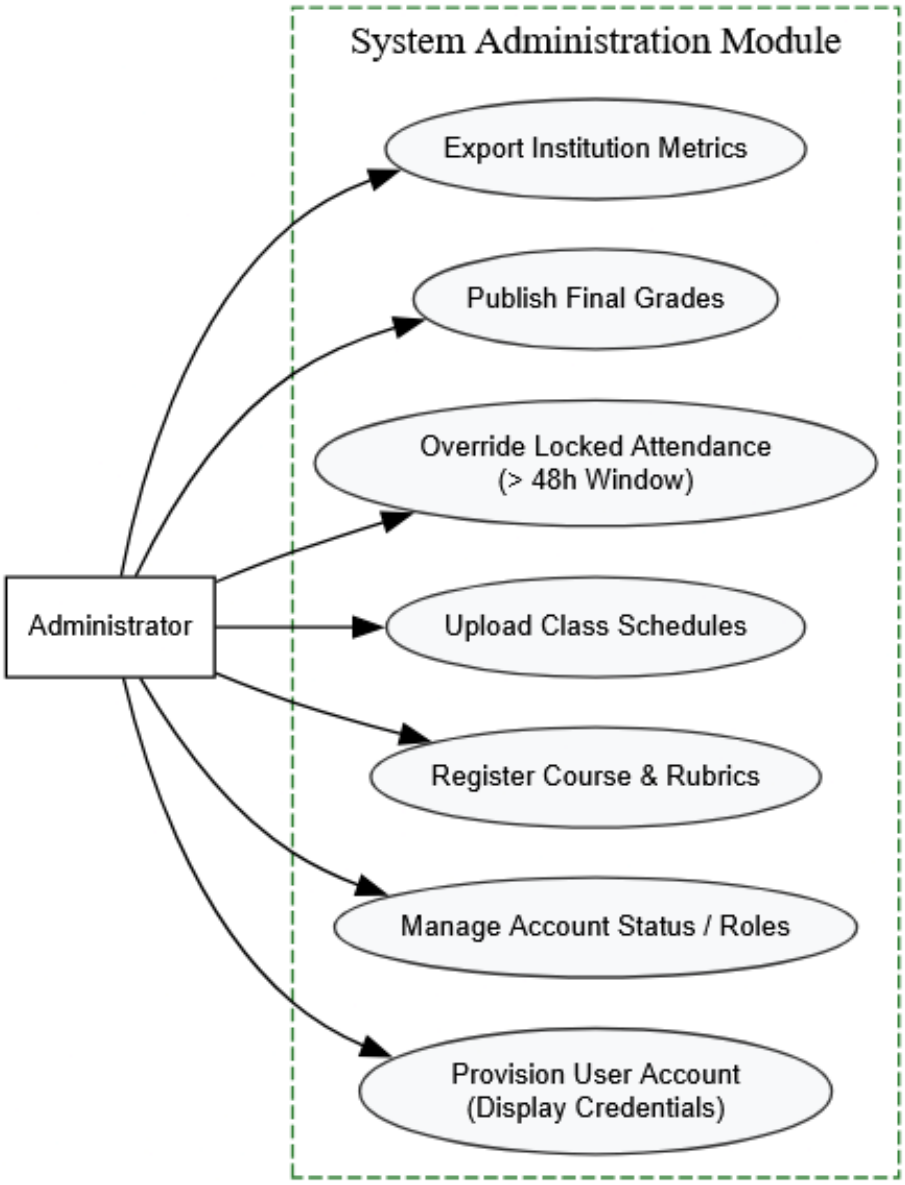



Figure 5: System Administration Use Case Diagram

# 8 Requirements Traceability Matrix (RTM)

| Req ID | Requirement Summary | Design Spec Ref | Target Module | Test Case Ref | Status |
|---|---|---|---|---|---|
| **SYS-F-001** | Provision user accounts & display credentials | DS-ACC-01 | Account Module | TC-ACC-01 | Planned |
| **SYS-F-002** | Mandate password update on first login | DS-ACC-02 | Auth Module | TC-ACC-02 | Planned |
| **SYS-F-003** | Update personal contact details | DS-ACC-03 | Profile Module | TC-ACC-03 | Planned |
| **SYS-F-004** | Manage account metadata and status | DS-ACC-04 | Admin Module | TC-ACC-04 | Planned |
| **SYS-F-005** | Register course & set rubric scheme | DS-CRS-01 | Course Module | TC-CRS-01 | Planned |
| **SYS-F-006** | Student self-enrollment (max 6 courses) | DS-CRS-02 | Enrollment Module | TC-CRS-02 | Planned |
| **SYS-F-007** | Create and publish class schedules | DS-CRS-03 | Schedule Module | TC-CRS-03 | Planned |
| **SYS-F-008** | Restrict faculty roster visibility | DS-CRS-04 | RBAC / Course | TC-CRS-04 | Planned |
| **SYS-F-009** | Digitized binary roll-call attendance | DS-ATT-01 | Attendance Module | TC-ATT-01 | Planned |
| **SYS-F-010** | 48-hour automated attendance freeze | DS-ATT-02 | Attendance Module | TC-ATT-02 | Planned |
| **SYS-F-011** | Administrative attendance override | DS-ATT-03 | Admin / Attendance | TC-ATT-03 | Planned |

| Req ID | Requirement Summary | Design Spec Ref | Target Module | Test Case Ref | Status |
|---|---|---|---|---|---|
| **SYS-F-012** | Student attendance dashboard view | DS-ATT-04 | Dashboard Module | TC-ATT-04 | Planned |
| **SYS-F-013** | Record rubric marks in `Draft` state | DS-GRD-01 | Grading Module | TC-GRD-01 | Planned |
| **SYS-F-014** | Submit draft grades with weight total | DS-GRD-02 | Grading Module | TC-GRD-02 | Planned |
| **SYS-F-015** | Publish final grades to students | DS-GRD-03 | Admin / Grading | TC-GRD-03 | Planned |
| **SYS-F-016** | Submit regrade dispute (`Under Review`) | DS-GRD-04 | Dispute Module | TC-GRD-04 | Planned |
| **SYS-F-017** | Resolve dispute and re-publish grade | DS-GRD-05 | Grading Module | TC-GRD-05 | Planned |
| **SYS-F-018** | Student consolidated GPA dashboard | DS-REP-01 | Dashboard Module | TC-REP-01 | Planned |
| **SYS-F-019** | Faculty class summary analytics | DS-REP-02 | Analytics Module | TC-REP-02 | Planned |
| **SYS-F-020** | Admin institution-level metrics export | DS-REP-03 | Analytics Module | TC-REP-03 | Planned |
| **SYS-NF-001** | Latency $< 800$ ms at 95th percentile | DS-PERF-01 | System-wide | TC-PERF-01 | Planned |
| **SYS-NF-002** | 200 concurrent transaction capability | DS-PERF-02 | Database Layer | TC-PERF-02 | Planned |

| Req ID | Requirement Summary | Design Spec Ref | Target Module | Test Case Ref | Status |
|---|---|---|---|---|---|
| **SYS-NF-003** | 99.0% operational availability | DS-OPS-01 | Infrastructure | TC-OPS-01 | Planned |
| **SYS-NF-004** | Responsive UI (375px-1920px, $\leq 3$ clicks) | DS-UX-01 | Web Presentation | TC-UX-01 | Planned |
| **SYS-NF-005** | 3-tier architecture & $\geq 70\%$ test cover | DS-QUAL-01 | Architecture | TC-QUAL-01 | Planned |
| **PRJ-SO-001** | Data isolation and RBAC objective | DS-SEC-01 | System-wide | TC-SEC-04 | Planned |
| **PRJ-SO-002** | Workflow state integrity objective | DS-SEC-02 | Core Business Rules | TC-ATT-02, TC-GRD-03 | Planned |
| **PRJ-SR-001** | Mandatory TLS 1.3 encryption | DS-SEC-03 | Network Layer | TC-SEC-01 | Planned |
| **PRJ-SR-002** | Bcrypt password hashing (factor 12) | DS-SEC-04 | Auth Module | TC-SEC-02 | Planned |
| **PRJ-SR-003** | 48-hour HTTP-only JWT sessions | DS-SEC-05 | Auth Middleware | TC-SEC-03 | Planned |
| **PRJ-SR-004** | Endpoint RBAC enforcement middleware | DS-SEC-06 | API Middleware | TC-SEC-04 | Planned |
| **PRJ-SR-005** | ORM parameterization (SQLi/XSS prevention) | DS-SEC-07 | Data Access Layer | TC-SEC-05 | Planned |